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Electronic Supplementary Material for the paper "Joint multizonal transdimensional Bayesian inversion of surface wave dispersion and ellipticity curves for local near-surface imaging"

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SUPPLEMENT S1: CONVERGENCE OF THE MCMC SAMPLING PROCEDURE

The purpose of the Markov chain Monte-Carlo (MCMC) sampling algorithm is to collect draws representing a discrete approximation of the posterior PDF. Such a sampling procedure should be treated carefully as it might be unrepresentative in some cases (e.g., Gelman et al. 2013, pp. 281-285). The first issue is a dependence of the early chain steps on an initial model. It is usually overcome by discarding early samples within a user-defined burn-in period, after which the sampling procedure converged to a stationary state. The second issue is that a single chain of steps possesses a within-sequence correlation of samples, which may be overcome by using multiple parallel Markov chains. Anyway, assessing convergence and monitoring of the sampling procedure should not be neglected when using the MCMC sampling algorithm.

We use the Parallel Tempering technique (Sambridge 2014) to suppress the within-sequence correlation of samples. In such a technique, the sampling is performed by multiple parallel Markov chains (cold sampling and hot exploration chains). These may swap their assigned temperatures, while the sample draws origin from the cold sampling chains only. To tackle the issue of a dependence on an initial model, we use random and independent initial models for all the parallel chains, and we are discarding early samples within the burn-in period. The remaining question is whether the sampling procedure converged to a stationary state. Assessing convergence may be performed by monitoring all model parameters in the multiple Markov chains having initial models dispersed throughout model space. When the distribution of a single chain is close to the distribution of all the chains mixed together, it can be approximating the target posterior PDF (e.g., Gelman et al. 2013, pp. 281-282). This is, unfortunately, an uneasy task in the transdimensional formulation with a varying number of model parameters.

To assess a convergence, we are monitoring the data variance reduction (Eq. (8)) on multiple computational nodes. One node has assigned one sampling and 19 exploration Markov chains, and it communicates with others through the Message Passing Interface (i.e., MPI node). Hence, one MPI node process 20 chains while producing one sequence of drawn samples. We deploy multiple parallel MPI nodes to increase the number of drawn samples and enlarge a diversity of initial models (e.g., 16, 24, or 48). In this supplement, we show the convergence of the data variance reduction for the single-zone synthetic test (Fig. S1.1), multizonal synthetic test (Fig. S1.2), and the real data inversion (Fig. S1.3). Each sequence (distinguished by color) consists of the highest data variance reduction of proposed models per MPI node. To assess the convergence, we may compare the densely sampled distribution of multiple sequences. When the distribution of a single sequence is close to all the sequences mixed together, its sampling chain can be approximating the target posterior PDF. We note that there are some sparse cases of sudden temporary declines in the data variance reduction. These are because each sequence results from one sampling and 19 exploration chains, further complicated by numerous temperature swaps. Hence, these outliers are related to the exploration chains that have a higher dispersion; nevertheless, they do not interfere with the convergence of the sampling chain. A comparison of Figs S1.1 and S1.2 demonstrates that the convergence is faster in the multizonal than single-zone inversion. And Fig. S1.3 shows a rapid convergence in the inversion of the real data measured at the

SENGL site. To conclude, our sampling procedure may be stationary after approximately 3,000-10,000 burn-in steps (the optimum length of the burn-in period varies from case to case).

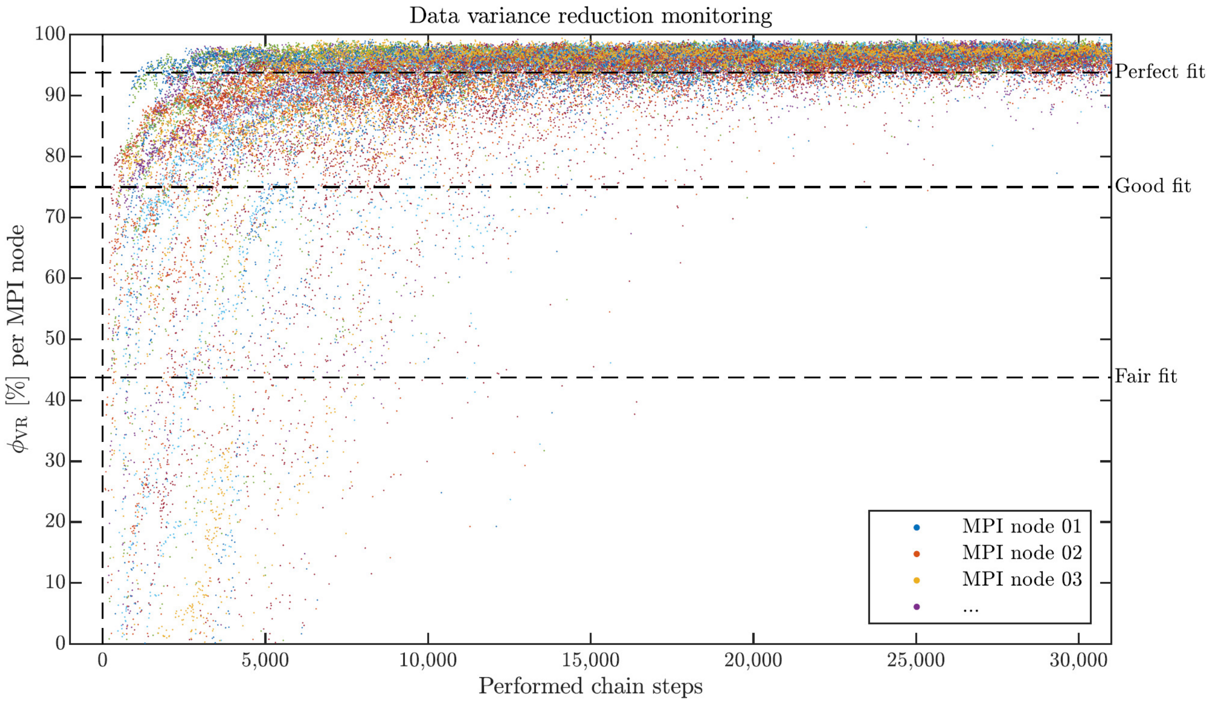

Figure S1.1. Convergence of the MCMC sampling procedure: Inversion of the synthetic data with the single-zone transdimensional model space (Fig. 5 in the main text). There were deployed 24 MPI nodes processing 20 parallel Markov chains per each. X-axis shows number of performed chain steps (0 = initial models). Y-axis shows maximal data variance reduction of proposed models per each MPI node.

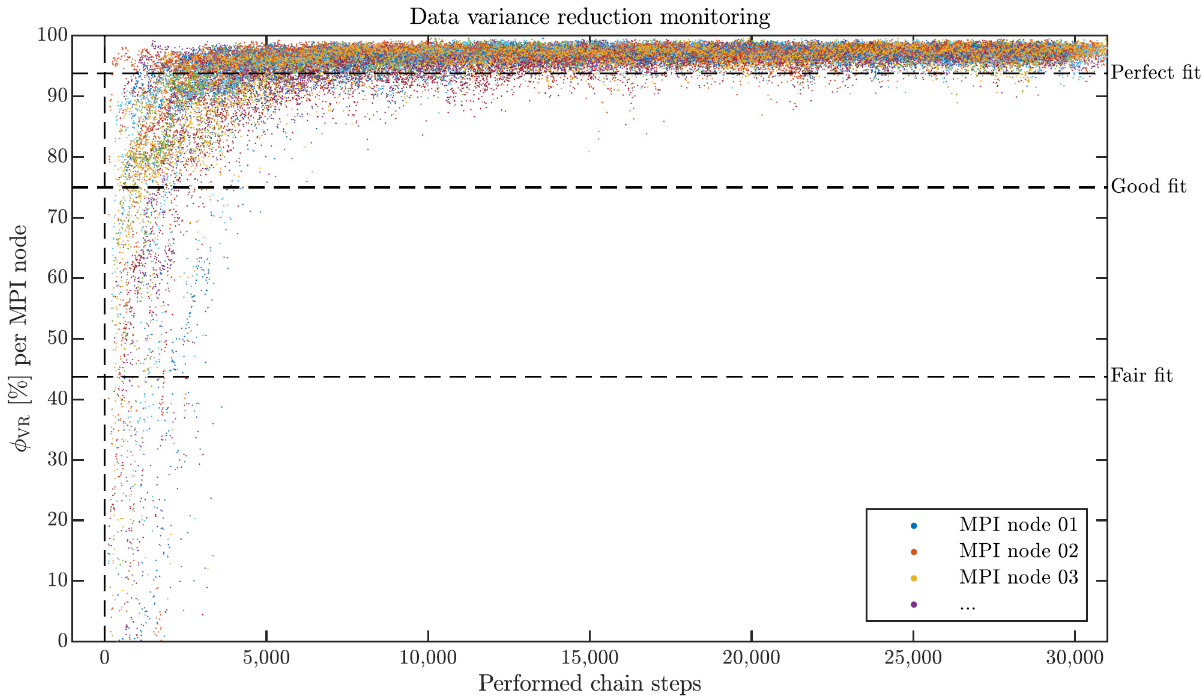

Figure S1.2. Convergence of the MCMC sampling procedure: Inversion of synthetic data with the multizonal transdimensional model space (Fig. 7 in the main text). There were deployed 24 MPI nodes processing 20 parallel Markov chains per each. X-axis shows number of performed chain steps (0 = initial models). Y-axis shows maximal data variance reduction of proposed models per each MPI node.

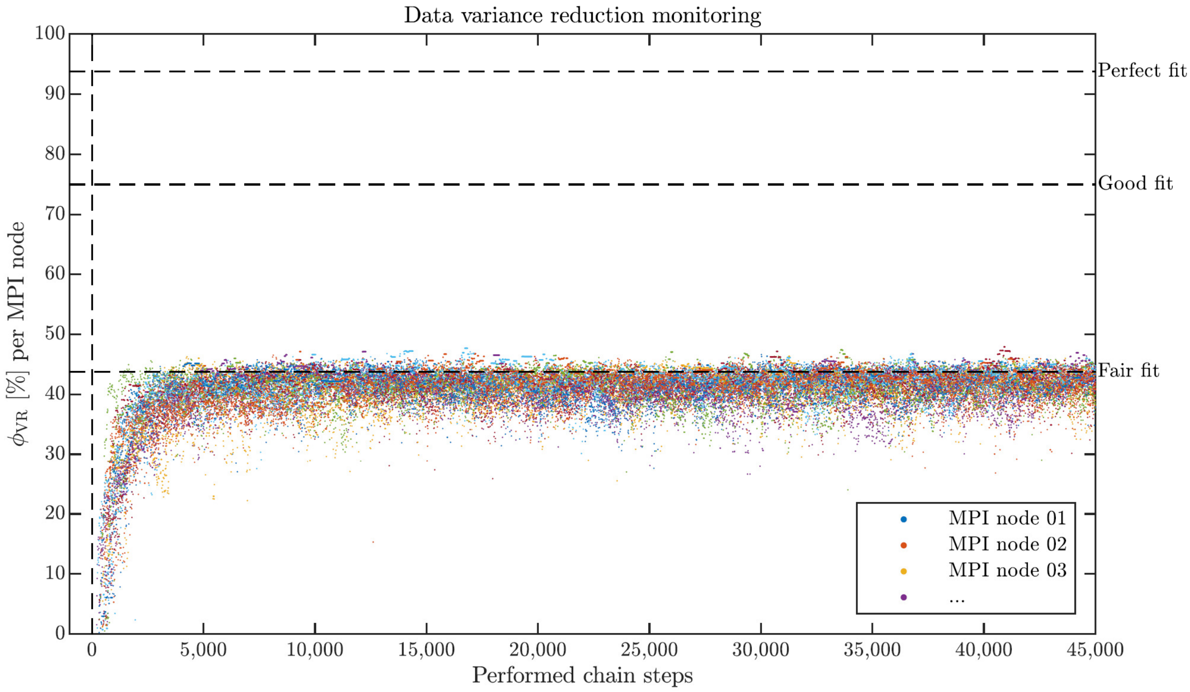

Figure S1.3. Convergence of the MCMC sampling procedure: Inversion of the real data measured at the SENGL site (Fig. 13 in the main text). There were deployed 16 MPI nodes processing 20 parallel Markov chains per each. X-axis shows number of performed chain steps (0 = initial models). Y-axis shows maximal data variance reduction of proposed models per each MPI node.

SUPPLEMENT S2: RANDOM WALK WITH THE HOMOGENEOUS PRIOR

In our approach, the prior PDF of a parameter a is efficiently homogeneous within an interval of physically plausible values $A = [a^{\min}, a^{\max}]$. This is a weakly informative and proper prior that can be expressed by Eq. (11) of the main text. The respective continuous parameter space is explored by a random walk utilizing the Gaussian distribution as the proposal distribution of random steps. The Gaussian proposal distribution is given by

$$q(a'|a) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{a'-a}{\sigma}\right)^2\right) \tag{S2.1}$$

where a' is the proposed parameter value, and σ is the standard deviation of the Gaussian proposal distribution that is considered rather small with respect to A. The proposal distribution is re-centered at each step of the chain to the current parameter value a. The Gaussian distribution may randomly propose a value outside the interval A as it is non-zero over the entire $\mathbb{R}$. Such a proposed value may be rejected immediately or generated again, but it leads to a higher rejection rate or a biased prior PDF close to the edges of the interval A, respectively. Hence, we utilize the first-order mirroring of this symmetrical proposal distribution at edges of A, following Hallo & Gallovič (2020), as follows

$$a'' = \begin{cases} 2a^{\min} - a' & \text{if } a' < a^{\min} \\ 2a^{\max} - a' & \text{if } a' > a^{\max} \\ a' & \text{if } a' \in A \end{cases} \tag{S2.2}$$

where a'' is the new proposed parameter value. Then, a chain of steps produces uniformly distributed random samples drawn from A without rejections due to the vicinity of the interval edge. An example of the performance of such an algorithm is shown in Fig. S2.1 that is produced by the Matlab program below.

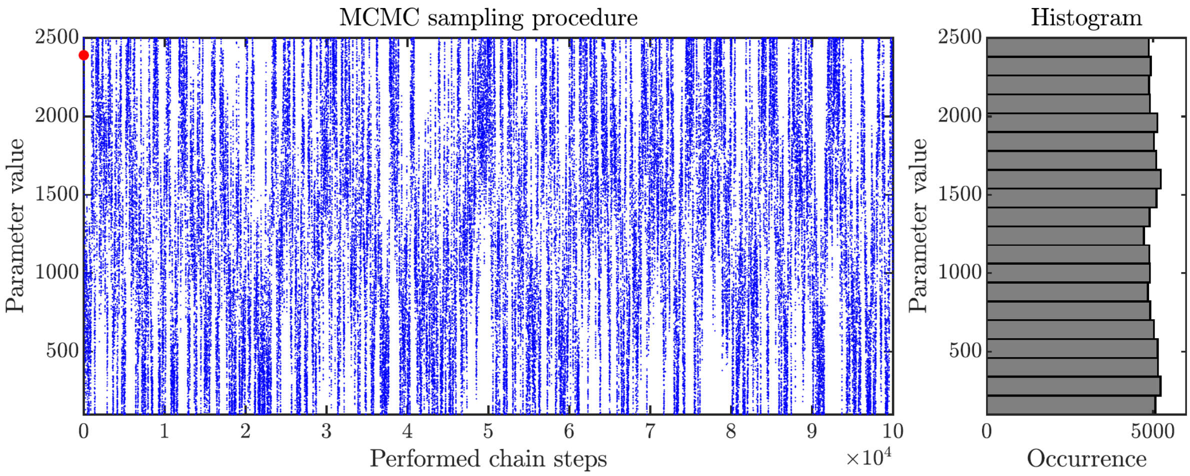

Figure S2.1. An example of the Gaussian random walk in the parameter space with the efficiently homogeneous prior. The initial random value is shown by the red dot. This figure is produced by the Matlab code below.

```
% Performance test of the implemented MCMC random walk
% Programmed by: Miroslav Hallo, Swiss Seismological Service, ETH Zurich.
% Hallo, M., Imperatori, W., Panzera, F., Fäh, D. (2021). Joint multizonal
%      transdimensional Bayesian inversion of surface wave dispersion and
%      ellipticity curves for local near-surface imaging, Geophys. J. Int.
% Version: Tested on the Matlab R2019b
% This code is published under the GNU Public License for non-commercial purposes.
% It is distributed in the hope that it will be useful, but without any warranty.
% -----------------------------------------------------------------
% MCMC settings
amin = 100;   % Lower threshold of the parameter value (homogeneous prior)
amax = 2500;  % Upper threshold of the parameter value (homogeneous prior)
step = 100;   % Standard deviation of the Gaussian proposal distribution
N = 1000000;  % Number of performed MCMC steps
% -----------------------------------------------------------------
% MCMC sampling procedure
% Allocate vector for N samples
m = zeros(1,N);
% Initial random model (drawn from the homogeneous prior)
m(1) = amin+(rand(1,1)*(amax-amin));
% Start of the sampling procedure
for s = 2:N
    % Propose a new model (using Gaussian proposal distribution)
    mProp = m(s-1)+randn(1,1)*step;
    % Mirroring of the proposal distribution at edges; Hallo and Gallovic (2020)
    if mProp>amax; mProp=amax-(mProp-amax); end
    if mProp<amin; mProp=amin+(amin-mProp); end
    % Metropolis-Hastings acceptance alfa=p(d|m')/p(d|m)
    alfa = 1; % << homogeneous prior, monitoring of the random walk only
    % Accept or reject the proposed model? (accepts all in this case)
    if rand(1,1)<alfa; m(s) = mProp; else; m(s) = m(s-1); end
end
% -----------------------------------------------------------------
% Plot the sampling procedure
figure('Units','normalized','Position',[0.1,0.1,0.8,0.8])
subplot(1,4,1:3)
plot(m,'.b','MarkerSize',1); hold on;
plot(m(1),'.r','MarkerSize',13); hold off;
set(gca,'Ylim',[amin amax])
xlabel('Performed chain steps'); ylabel('Parameter value')
title('MCMC sampling procedure')
subplot(1,4,4)
Nbins = max(5,abs(N/5000));
edges = [amin, amin+((1:Nbins)*(amax-amin)/Nbins)];
histogram(m,edges,'EdgeColor','k','FaceColor',[0.5 0.5 0.5],'FaceAlpha',1)
set(gca,'Xlim',[amin amax])
xlabel('Parameter value'); ylabel('Occurrence')
view(90,-90)
title('Histogram')
return
```

SUPPLEMENT S3: REVERSIBLE JUMP WITH THE RECIPROCAL PRIOR

The parameterization with a varying number of layers leads to the formulation of the model space by Green (1995, 2003). In this formulation, we assume to have a countable collection of model states $\mathcal{K}$ indexed by $k \in \mathcal{K}$, which differ from each other by the number of layers. Each model space state k has its own n_k-dimensional vector space $\mathcal{R}^{n_k}$, and the union state-space $\mathcal{X}$ for across-model simulations is defined following Green (2003) as

$$\mathcal{X} = \cup_{k \in \mathcal{K}}(\{k\} \times \mathcal{R}^{n_k}) \quad \text{(S3.1)}$$

The union state-space $\mathcal{X}$ can be explored by the reversible jump Markov chain Monte Carlo algorithm (rjMCMC, Green 1995). Green (1995) showed that the rjMCMC converges to a discrete approximation of the posterior PDF, if the Metropolis-Hastings acceptance probability α is given by

$$\alpha(\mathbf{m}'|\mathbf{m}) = \min\left(1, \frac{p(\mathbf{m}')}{p(\mathbf{m})} \frac{p(\mathbf{d}_{\text{obs}}|\mathbf{m}')}{p(\mathbf{d}_{\text{obs}}|\mathbf{m})} \frac{q(\mathbf{m}|\mathbf{m}')}{q(\mathbf{m}'|\mathbf{m})} |\mathbf{J}|\right) \quad \text{(S3.2)}$$

where $\mathbf{m} = (k, \mathbf{w}_k)$ is the current model, $\mathbf{m}' = (k', \mathbf{w}'_{k'})$ is the proposed model, $p(\mathbf{m})$ is the prior of model parameters, $p(\mathbf{d}_{\text{obs}}|\mathbf{m})$ is the likelihood function, and $|\mathbf{J}|$ is determinant of the Jacobian of the model space transformation. Terms $q(\mathbf{m}'|\mathbf{m})$ and $q(\mathbf{m}|\mathbf{m}')$ are proposal distributions of the forward and reverse steps, respectively (e.g., Sambridge et al. 2006). Model parameters are continuous variables (see Supplement S2), while the model space state k is a discrete property ($k \in \mathbb{Z}^+$, where $k \propto \dim(\mathbf{m})$ in our case). It is convenient to separate the prior conditioned by a given k and the prior of the state itself from Eq. (S3.2) as follows

$$\alpha(\mathbf{m}'|\mathbf{m}) = \min\left(1, \frac{p(k')}{p(k)} \frac{p(\mathbf{w}'_{k'}|k')}{p(\mathbf{w}_k|k)} \frac{p(\mathbf{d}_{\text{obs}}|\mathbf{m}')}{p(\mathbf{d}_{\text{obs}}|\mathbf{m})} \frac{q(\mathbf{m}|\mathbf{m}')}{q(\mathbf{m}'|\mathbf{m})} |\mathbf{J}|\right) \quad \text{(S3.3)}$$

where $p(k)$ is the prior probability of the current state k, $p(k')$ is the prior probability of the proposed state k', $p(\mathbf{w}_k|k)$ and $p(\mathbf{w}'_{k'}|k')$ are prior PDFs of model parameters given states k and k', respectively (e.g., Hallo & Gallovič 2020). We could also separate the proposal distributions q; however, it is not necessary in our case as we use the birth-death algorithm with an equal probability of the birth and death proposals, i.e. $Q_{\text{B}}(k+1|k) = Q_{\text{D}}(k|k+1)$. As is shown in Eq. (S3.3), we can prescribe prior requirements on the model space states through $p(k)$.

The number of layers can be treated as a measure of the subsurface model complexity. Indeed, models with fewer layers are simpler and leave less space for a potential overfitting of the observed data. On the other hand, too few layers may cause an underfitting (i.e., a bias-variance tradeoff). Hence, we a priori prefer models with fewer layers that still fit the data. From the point of view of the Bayesian theory, a prior in favor of simpler models is not necessarily required as a model with fewer layers will automatically have an enhanced (sharper) posterior PDF. This is an objective Occam's razor as a direct consequence of using a Bayesian method (Jefferys & Berger 1992; Malinverno 2002). Note that this is the reason why MAP models have a lower number of layers than ML models. However, the posterior marginal PDF of the transdimensional method is a composite from multiple model spaces (see Eq. (S3.1)) that have different parameters and inter-parameter correlations. Such a posterior marginal PDF can be used for an objective interpretation in marginal dimensions (i.e., v_{S}, v_{P}, and ρ), but it may introduce a bias in the inter-parameter correlations due to a mixture of multiple model space states. Moreover, the joint inversion exhibits significant inherent non-uniqueness that may complicate the posterior PDF by the presence of multiple posterior maxima. In other words, a standard joint transdimensional inversion can infer objective values of posterior seismic velocities, not the layered profiles itself. To this end, we prescribe an explicit Occam's razor for the number of layers through the prior of the model space states $p(k)$. This is a law of parsimony where the layered profile samples are as simple as possible but not simpler than required by the observed data (see Appendix B). In particular, following Hallo & Gallovič (2020), we use the discrete reciprocal distribution expressed as follows

$$p(k) = c_{\mathcal{K}} k^{-1} \quad \text{(S3.4)}$$

where $c_{\mathcal{K}}$ is a normalization constant. The prior probability $p(k)$ in Eq. (S3.4) is a proper prior because k has a positive integer value ($k \in \mathbb{Z}^+$), and it is indexing the countable collection of model states $\mathcal{K}$ as required by the definition of the union state-space $\mathcal{X}$ (Green 2003). Then, a substitution of the first term of the Metropolis-Hastings acceptance by Eq. (S3.4) has one of the following three forms

$$\frac{p(k')}{p(k)} = \frac{p(k)}{p(k)} = 1 \tag{S3.5}$$

$$\frac{p(k')}{p(k)} = \frac{p(k+1)}{p(k)} = \frac{c_{\mathcal{K}}(k+1)^{-1}}{c_{\mathcal{K}}k^{-1}} = \frac{k}{k+1} \tag{S3.6}$$

$$\frac{p(k')}{p(k)} = \frac{p(k-1)}{p(k)} = \frac{c_{\mathcal{K}}(k-1)^{-1}}{c_{\mathcal{K}}k^{-1}} = \frac{k}{k-1} \tag{S3.7}$$

Substitutions in Eqs. (S3.5), (S3.6) and (S3.7) are relevant to the M-H acceptance probability of the perturb, birth, and death moves of the birth-death rjMCMC, respectively. These terms originate from the explicit prior of the model space states; hence, they are beyond the prior of model parameters (that cancel as shown in Appendix A). As a consequence, the M-H acceptance of the birth or death move prioritize lower k (i.e., the lower number of layers).

For a deeper understanding, let's imagine two models $\mathbf{m}_1$ and $\mathbf{m}_2$ having different number of layers with $\dim(\mathbf{m}_2) > \dim(\mathbf{m}_1)$. These models are arbitrary but they have an equal fit to the observed data that can be expressed by

$$\frac{p(\mathbf{d}_{\text{obs}}|\mathbf{m}_2)}{p(\mathbf{d}_{\text{obs}}|\mathbf{m}_1)} = \frac{p(\mathbf{d}_{\text{obs}}|\mathbf{m}_1)}{p(\mathbf{d}_{\text{obs}}|\mathbf{m}_2)} = 1 \tag{S3.8}$$

So, one would prefer the model $\mathbf{m}_1$ as it is simpler. Next, imagine the M-H acceptance in the case of the birth-from-prior technique without preference of model space states, i.e. $p(k')/p(k) = 1$. Then, the M-H acceptance probabilities of the birth and death moves are mathematically equal

$$\alpha_{\text{B}}(\mathbf{m}_2|\mathbf{m}_1) = \alpha_{\text{D}}(\mathbf{m}_1|\mathbf{m}_2) = 1 \tag{S3.9}$$

This may lead to a surplus of unnecessary layers in the sampling velocity profiles, even though the posterior marginal PDF of v_{S} will be sharper in the case of $\mathbf{m}_1$ (because the prior PDF of $p(\mathbf{m}_1) > p(\mathbf{m}_2)$). On the other hand, in the case of the rjMCMC with the explicit prior of states, i.e. Eqs (13) and (14) in the main text, the M-H acceptance probabilities prioritize the model with fewer layers:

$$\alpha_{\text{B}}(\mathbf{m}_2|\mathbf{m}_1) < \alpha_{\text{D}}(\mathbf{m}_1|\mathbf{m}_2) \tag{S3.10}$$

while other characteristics are preserved. This is the essence of our implemented law of parsimony.

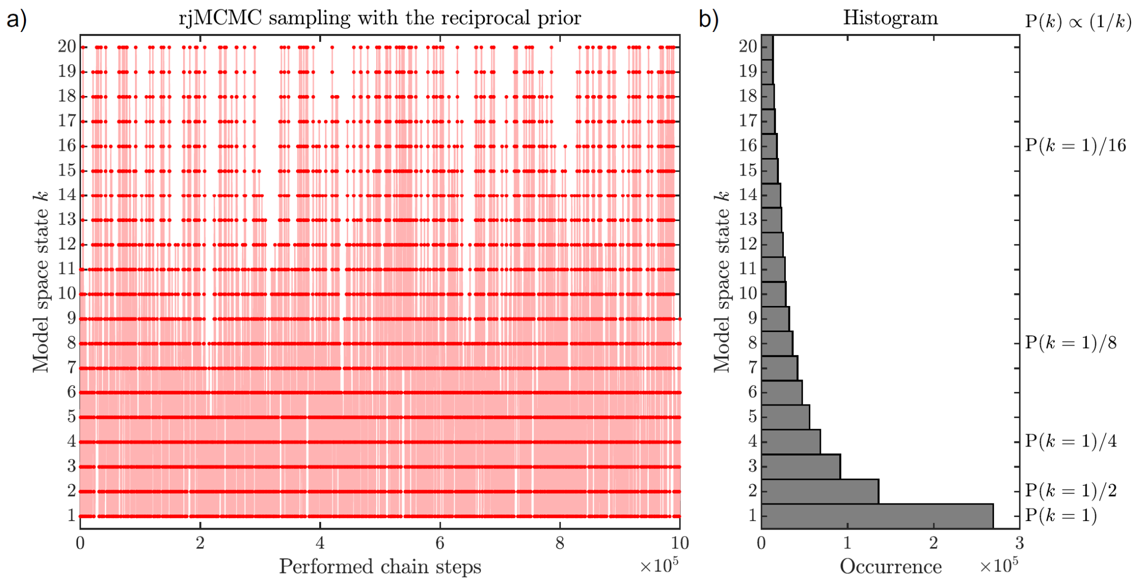

Figure S3.1. A random exploration of the union state space by the reversible-jump MCMC algorithm with the reciprocal prior. Panel a) shows random jumps between neighboring discrete model space states during the sampling procedure. Panel b) shows the occurrence of the individual discrete states in the sampling chain. Indeed, the latter shows the prescribed reciprocal distribution. This figure is produced by the Matlab code below.

In this supplement, we provide a simple Matlab program of the reversible-jump algorithm with the implemented reciprocal prior. The output of the program, shown in Fig. S3.1, demonstrates that the prior of the model space states $p(k)$ is truly the prescribed reciprocal distribution of Eq. (S3.4). In particular, see the decline of the probability (i.e., the normalized occurrence) with increasing k (k is proportional to the number of layers). Such an algorithm can be used if the physical complexity of the model increases proportionally to the number of dimensions of the model vector **m**.

```
% Simple test of the reversible-jump algorithm with the reciprocal prior
% Programmed by: Miroslav Hallo, Swiss Seismological Service, ETH Zurich.
% Hallo, M., Imperatori, W., Panzera, F., Fäh, D. (2021). Joint multizonal
%      transdimensional Bayesian inversion of surface wave dispersion and
%      ellipticity curves for local near-surface imaging, Geophys. J. Int.
% Version: Tested on Matlab R2019b
% This code is published under the GNU Public License for non-commercial purposes.
% It is distributed in the hope that it will be useful, but without any warranty.
% ------------------------------------------------------------------
% rjMCMC settings
Prj = 0.05; % Probability of transdimensional proposals (P_birth=P_death)
kmax = 20; % Upper threshold of model space states (proportional to dimensions)
N = 1000000; % Number of performed rjMCMC steps
% ------------------------------------------------------------------
% Birth-death rjMCMC with the reciprocal prior
%    - implementation following Hallo and Gallovic (2020)
% Allocate vector for monitoring
kall = zeros(1,N);
% The model space state k of the initial model
kall(1) = 1;
% Start of the sampling procedure
for s = 2:N
    k = kall(s-1); % Current model space state k
    % Select move type (birth-death rjMCMC)
    challenge = rand(1,1);
    if challenge<Prj % Birth move (dim(m')>dim(m))
        kProp = k+1; % Proposed new state k'=k+1
        kProp = min(kmax,kProp); % Check the upper threshold
    elseif challenge<(2*Prj) % Death move (dim(m')<dim(m))
        kProp = k-1; % Proposed new state k'=k-1
        kProp = max(1,kProp); % Check the lower threshold
    else % Perturb move (dim(m')=dim(m))
        kProp = k; % Proposed new state k'=k
    end
    % Metropolis-Hastings acceptance with the homogeneous p(m) and
    %      reciprocal p(k)=const*(1/k)
    % Perturb move: alfa=(p(k')/p(k))*(p(d|m')/p(d|m))=p(d|m')/p(d|m)
    % Birth move: alfa=(p(k')/p(k))*(p(d|m')/p(d|m))=(k/(k+1))*(p(d|m')/p(d|m))
    % Death move: alfa=(p(k')/p(k))*(p(d|m')/p(d|m))=(k/(k-1))*(p(d|m')/p(d|m))
    alfa = (k/kProp)*1; % << monitoring of random jumps through states only
    %alfa = 1; % Test case WITHOUT preference of state
    % Accept or reject the proposed model?
    if rand(1,1)<alfa; kall(s) = kProp; else; kall(s) = k; end
end
% ------------------------------------------------------------------
% Plot the transdimensional sampling procedure
figure('Units','normalized','Position',[0.1,0.1,0.8,0.8])
subplot(1,3,1:2)
plot(kall,'or','MarkerSize',3);
plot(kall,'-o','Color',[1 .7 .7],'MarkerSize',1,'MarkerEdgeColor','r');
set(gca,'Ytick',1:kmax); set(gca,'Ylim',[0.5 kmax+0.5])
xlabel('Performed chain steps'); ylabel('Model space state k')
title('rjMCMC sampling with the reciprocal prior')
subplot(1,3,3)
histogram(kall,'EdgeColor','k','FaceColor',[0.5 0.5 0.5],'FaceAlpha',1)
set(gca,'Xtick',1:kmax); set(gca,'Xlim',[0.5 kmax+0.5])
xlabel('Model space state k'); ylabel('Occurrence')
view(90,-90)
title('Histogram')
return
```

SUPPLEMENT S4: DENSITY CONSTRAINED INVERSION OF THE REAL DATA

In the main text, we apply our inversion method to real data measured at the site of the SENGL seismic station (Fig. 12 in the main text). We use the single-zone transdimensional model space with the mass density within the interval of [1500, 2500] kg/m^3. The result of the inversion (Fig. 13 in the main text) implies that the inferred density profiles are unrealistic, and they may be considered purely as the algorithm's effort to reduce the misfit between real data and data modeled by a simple one-dimensional model.

In this supplement, we show an additional inversion test of the real data measured at the site of SENGL seismic station, using a narrow density profile of [1900, 2100] kg/m^3. All inversion settings and thresholds are the same as in the real data inversion of the main text. First, the data fit in Fig. S4.1 is almost identical to the fit in Fig. 12. It implies, that there is very low sensitivity on the mass density as discussed also in the main text. Next, the result of the inversion (Fig. S4.2) is similar to the inversion shown in Fig. 13. The posterior marginal PDFs of v_S and v_P are almost identical to the inversion in the main text. The marginal PDFs are slightly sharper as the volume of the model space has been reduced by constraining the mass density. The only difference is in the selected ML model that may be due to a smaller volume of the model space or a random incident. On the other hand, the posterior marginal PDF of the mass density differs in absolute values from the result in Fig. 13 due to different model space thresholds. This, once again, implies very low sensitivity on the mass density and that the inferred density profiles are unrealistic.

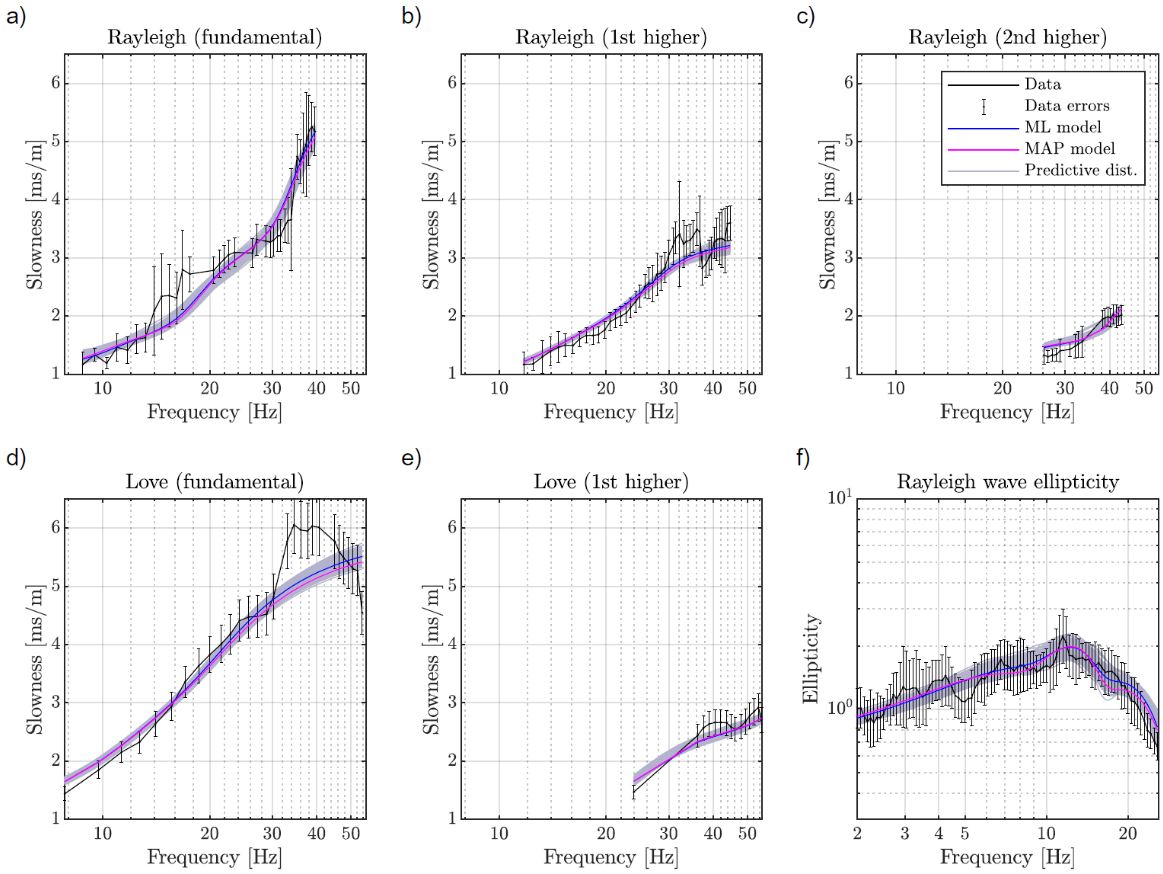

Figure S4.1. Fit of the real data measured at the SENGL site (black curves with errorbars) and data modeled in velocity models from the ensemble of solutions. Panels a) – e) show data retrieved by the MASW method (joint axes are used). Panel f) shows data retrieved by the RayDec method (Hobiger et al. 2009) from single-station measurements of ambient vibrations.

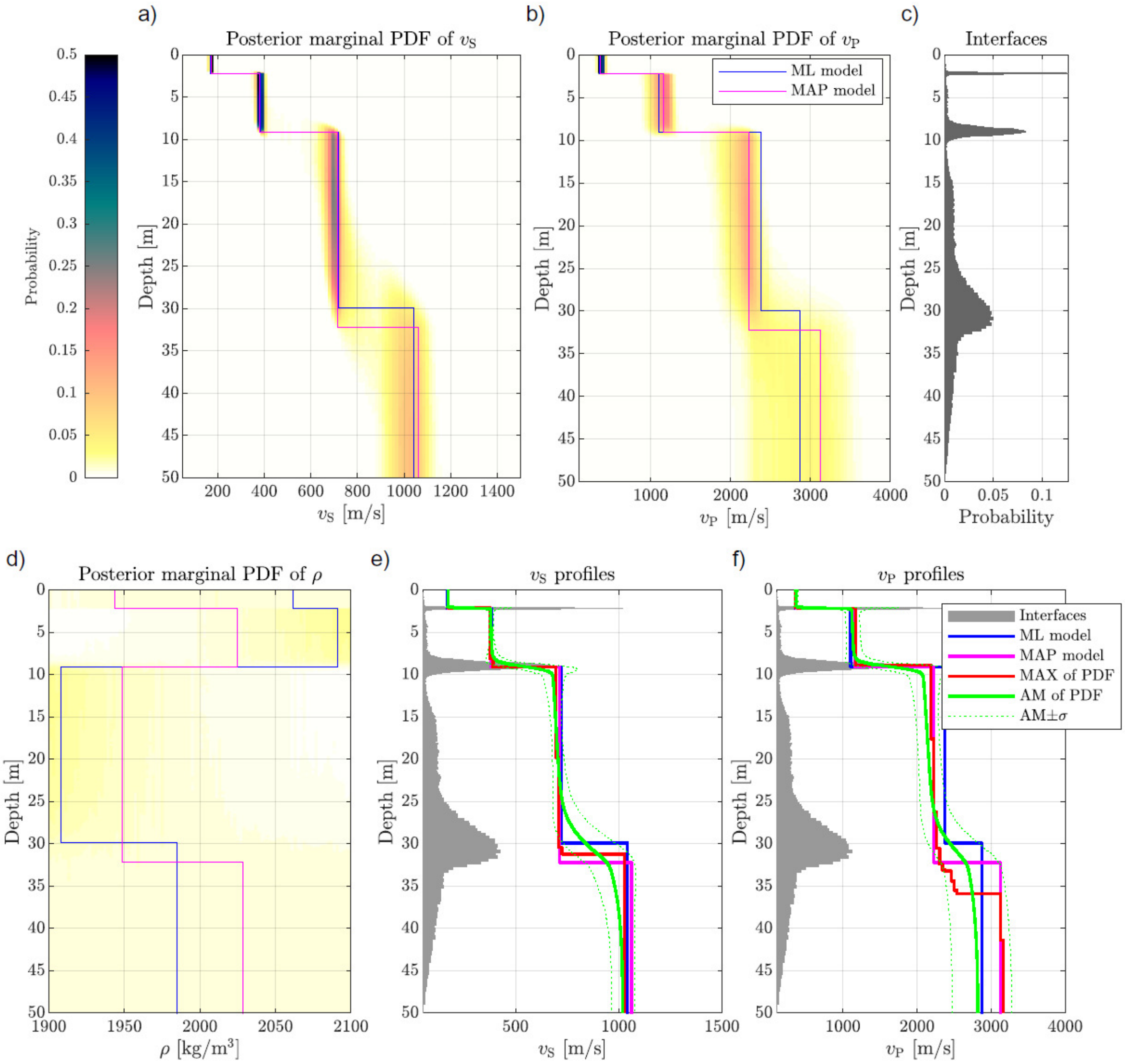

Figure S4.2. Inversion of the real data measured at the SENGL site using constrained mass density. Panels a), b), and d) show the posterior marginal PDF (colorscale) overlayed by the ML, MAP, and target models (see legend). Panel c) shows a histogram for the interface depth. Panels e) and f) show the inferred S-wave and P-wave velocity profiles, respectively.

SUPPLEMENT S5: RESIDUAL ANALYSIS OF THE REAL DATA INVERSION

The inversion of data measured at the SENGL site is a joint inversion of six different data types. The resultant fit between the modeled and observed data is shown by the posterior predictive distribution (Fig. 12) and qualitatively described by the data variance reduction (Fig. 14a). To further support the results, we examine here standardized residuals a posteriori (e.g., Dettmer et al. 2007). For the sake of the likelihood function in Eq. (6), we assume a Gaussian distribution of data errors centered on the observed values. A correlation of data errors is likely through all the used data types; nevertheless, these correlations are hardly predictable. Hence, the distribution of assumed data errors is approximated by a diagonal covariance matrix (i.e., data variance). Note that the variance of observed dispersion and ellipticity curves can be determined during processing of a raw field data; hence, the assumed variance of data can be empirical. To investigate how posterior distribution of errors differs from these assumptions, we compute a covariance matrix of standardized data residuals. The standardized residuals are data misfits between the posterior predictive distribution and observed data in standardized units. The covariance matrix is then evaluated numerically using predicted standardized data from the whole ensemble of solutions.

The resultant covariance matrix of standardized residuals from the real data inversion is shown in Fig. S5.1. If the data were not biased or have correlated errors, this covariance matrix would be an identity matrix. But in this case, the variance of standardized residuals is not uniformly equal to one. This is due to the presence of correlations of residuals as shown by non-zero off-diagonal covariance. Still, the amount of positive and negative covariance values is balanced and the resultant variance is reasonable. It means that our joint inversion works well, but data errors are correlated. Next, there are some data samples with particularly large variance of standardized residuals. These are related to the biased observed data. In particular, data samples 115-120 are related to the biased observed data of the fundamental Love wave dispersion curve in the 33–41 Hz frequency range (Fig. 12d) being probably related to shallow (< 2 m) heterogeneities along the active seismic line. Still, the amount of such a biased data samples is not dominant in this case; and hence, the joint inversion provides a robust joint solution. To further improve the inversion results and posterior model uncertainty estimate, we might get rid of the identified biased observed data or assume a full data covariance matrix and perform the inversion again.

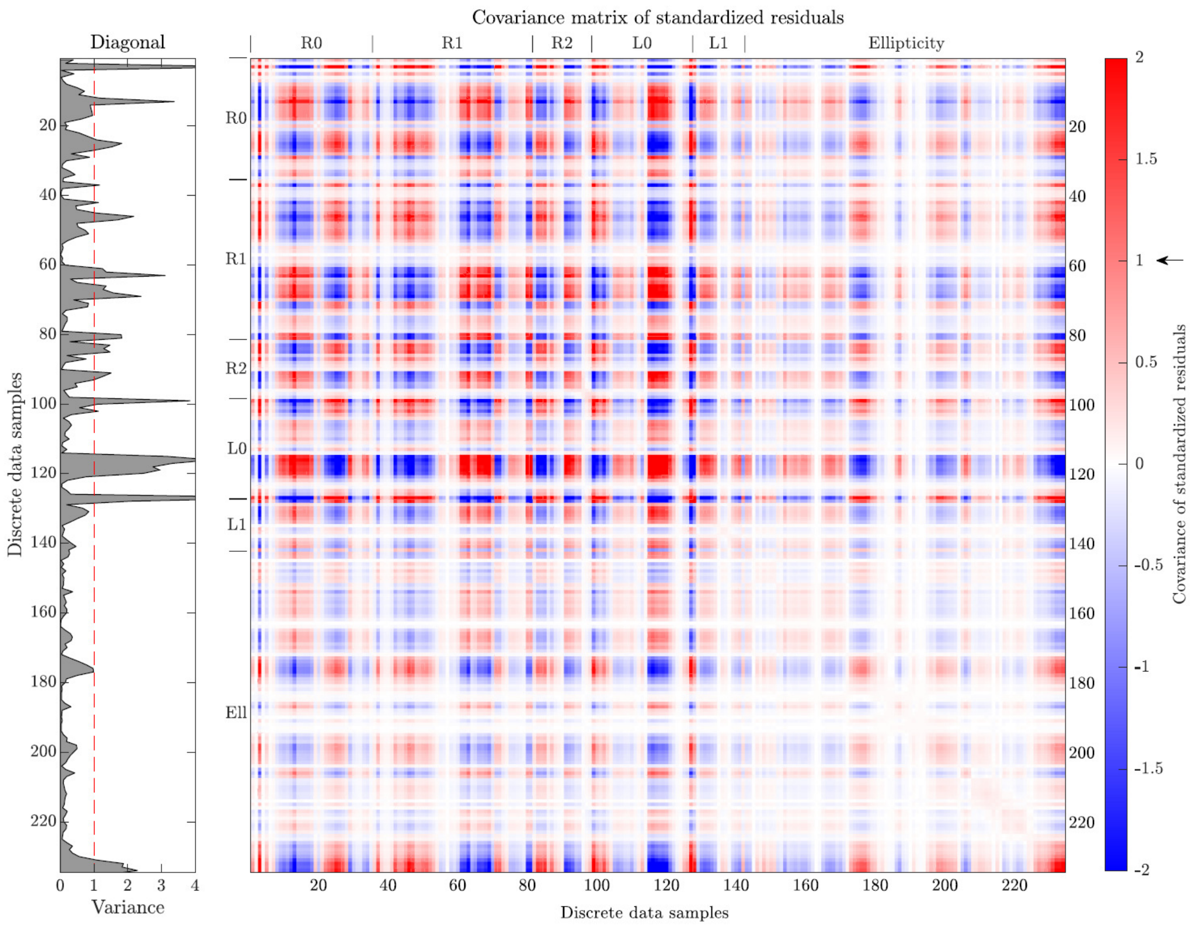

Figure S5.1. The covariance matrix of standardized residuals from the inversion of real data measured at the SENGL site (Fig. 12 in the main text). The standardized residuals are data misfits between the posterior predictive distribution and observed data in standardized units. Variances of standardized residuals are arranged along the descending diagonal. Data samples are discretized dispersion curves of R0, R1, R2 modes of Rayleigh waves, and L0, L1 modes of Love waves; and discretized ellipticity curve of Rayleigh waves. The frequency of discrete samples is increasing with their index within each data curve separately.